

Data Centers Impacts and Regulations

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1. Introduction to Data Centers

Data centers, schematically shown in Fig. 1, are large warehouse-like facilities to house organizations' IT infrastructure, including servers, storage systems, and networking equipment [1]. They are the backbone of model digital services and are essential for cloud computing, online transactions, streaming platforms, and artificial intelligence (AI). Although these large computing facilities have existed for decades, the number and capacity of data centers have expanded rapidly in recent years, especially with the growth of AI applications [2].

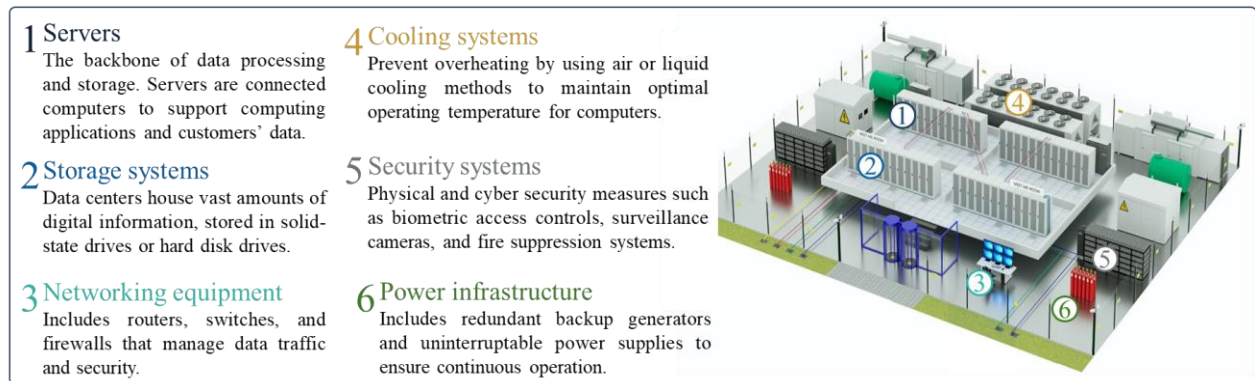


Fig. 1. Schematic illustration of a data center (adapted from [3]).

The United States, with more than 5,400 datacenters nationwide, has more data centers than the rest of the world combined. This number is expected to grow significantly, driven by more than \$2 trillion in projected global investment through 2030, nearly half of which is concentrated in the United States [2]. Fig. 2 shows some examples of these data centers in different states.



Fig. 2. Examples of data centers in different U.S. states.

Datacenters provide major economic and technological benefits, but they also raise important concerns. This report focuses on summarizing some of the potential negative impacts that data centers may create if their expansion is not fully planned, studied, and regulated. This report also reviews regulations and best practices introduced by federal, states, and private-sector organizations to support the safe, responsible, and beneficial integration of data centers into communities while minimizing their potential negative effects.

2. Data Center-Related Concerns

2.1. High Energy Consumption

Fig. 3 (a) shows the trend of data center electricity consumption over the past several years. From 2010 to 2018, their electricity use remained relatively stable. Between 2018 and 2022, their electricity consumption increased modestly as demand grew, while improvements in technology and infrastructure helped limit the rate of this increase. Since 2022, however, electricity consumption has risen sharply due to the rapid growth of large language models (LLMs), such as ChatGPT, and other AI-based platforms, which require larger and more energy-intensive data centers. Looking ahead, substantial uncertainty remains because the future scale of data centers, their actual demand, technological efficiency, and energy sources are still evolving. Even so, it is projected that data center operation could account for approximately 6.7% to 12% of total U.S. electricity demand in 2028 [4]. In other words, data center energy consumption has tripled over the past decade and could triple again by 2028 [5].

As shown in Fig. 3 (b), data centers are projected to consume a significant share of electricity in nearly all states by 2030, with some states (such as Virginia) expected to exceed 20% of total statewide electricity consumption [6]. It is also estimated that AI alone would drive a 165% increase in global data center electricity consumption by 2030 [2].

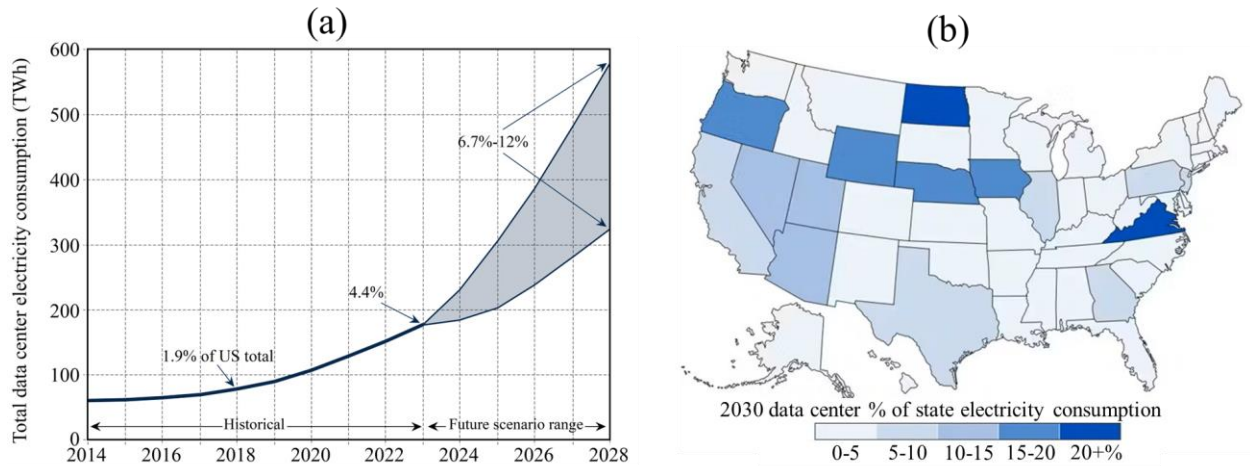


Fig. 3. U.S. data center electricity consumption (adapted from [4]); and (b) projected 2030 share of U.S. electricity consumption attributed to data centers (adapted from [6]).

As illustrated in Fig. 4, electricity consumption by U.S. data centers exceeds that of the rest of the world combined. Within the United States, the largest concentration of electricity consumption is in Virginia (at nearly 4,700 Megawatts), where the 2024 market nearly doubled compared with 2023 [7].

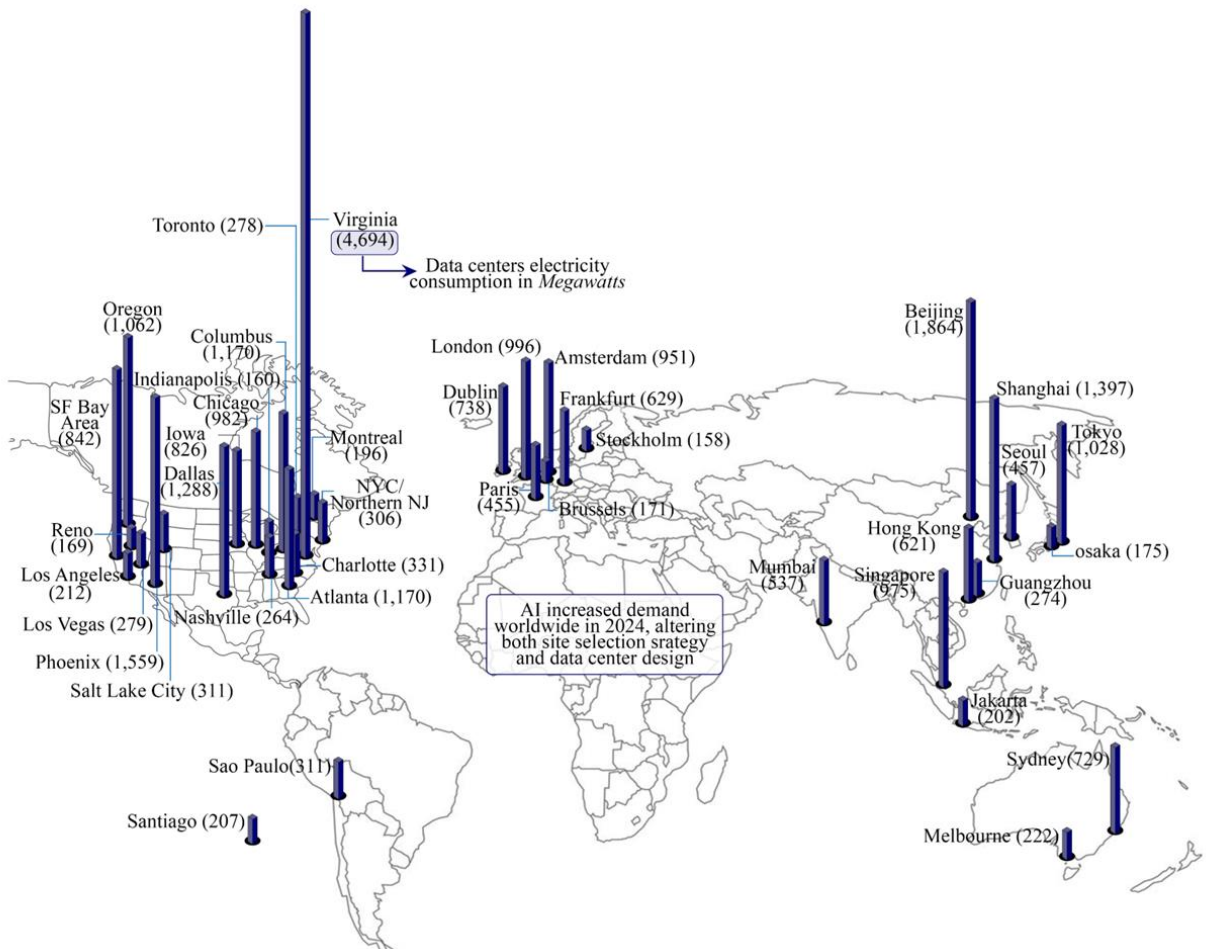


Fig. 4. Top 40 data center markets by electricity consumption in 2024 (adapted from [7]).

The source of electricity used by data centers is the major concern associated with their high energy demand. In general, this electricity is supplied in three main ways. The first is *grid interconnection*, in which a data center is connected to the same grid that serves other users, including residential and commercial customers. The second is *colocation with existing power plants*, where a data center is connected to an operating plant, such as an active nuclear power plant. This approach may require less transmission lines and supporting facilities, but it raises reliability concerns. The third is *colocation with new or restarted power plants*, in which a new plant is built, or an abandoned plant is restored, specifically to serve a nearby data center [5]. This report focuses on the grid interconnection approach because it is the most common one, although the other two approaches raise many of the same concerns. These concerns can be grouped into two main categories.

- a) **High and Variable Power Demand:** Data centers often have highly variable power consumption, which can affect power quality for other consumers. For example, Fig. 5 (a) shows the power demand of a Google data center during training an LLM, with fluctuations of more than 15 Megawatts over a short period of time. Such rapid fluctuations can stress grid infrastructure and reduce electricity reliability for other users [8]. Potential impacts include a reduced lifespan of household appliances, increased or even doubled peak demand, and a more unstable power experience for nearby communities. Fig. 5 (b) shows U.S. areas with the most distorted residential power experience based on readings from 770,000 home sensors in 2024. Two key findings emerge from this figure. First, more than 75% of highly distorted power readings occurred within 50 miles of significant data center activity. Second, the highest levels of power distortion were observed in areas adjacent to major concentrations of data centers, such as Loudoun County, Virginia (known as data center alley) and Columbus, Ohio (where 400 Megawatts of data center capacity was added in 2024, more than any other location in the country) [9].

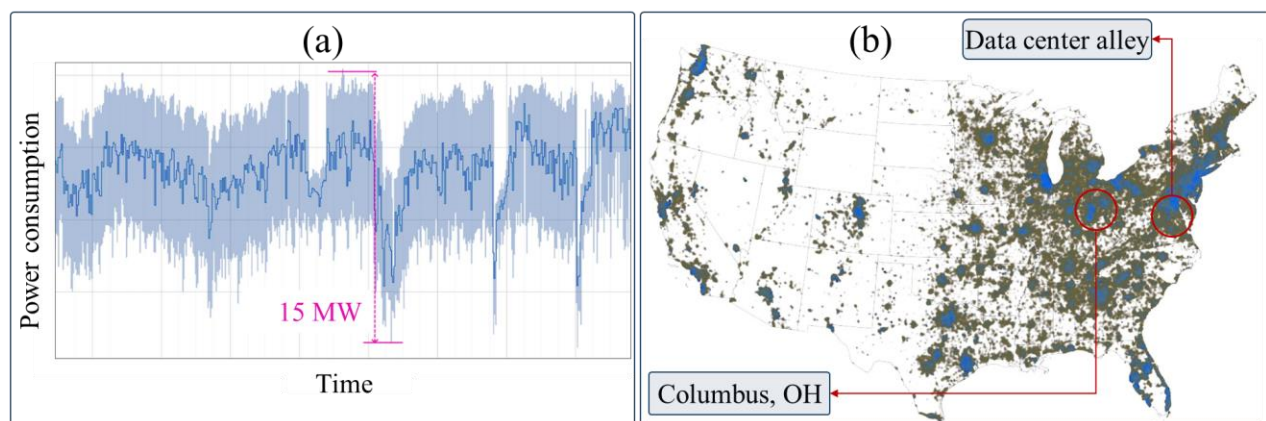


Fig. 5. Large power fluctuations in a data center (adapted from [8]); and b) areas with the most distorted power quality in 2024 (adapted from [9]).

- b) **Increased Costs:** Data centers can increase electricity costs in two main ways [1]. First, grid infrastructure, such as generation and transmission facilities, often must be upgraded when a new data center is built, and those costs are frequently subsidized, at least in part, by residents. Second, data centers add substantial new demand, making it difficult for electricity supply to keep pace, especially in the short term. This imbalance between supply and demand rates also contributes to higher electricity prices.

Fig. 6 (a) shows U.S. data centers power loads, with a sharp increase beginning around 2022, when large-scale data center expansion accelerated. As shown in Fig. 6 (b), this period also coincides

with a sharp increase in U.S. electricity prices. However, this was also the same period during which U.S. and global inflation rose sharply, and the Russia-Ukraine war disrupted global energy markets [10]. Therefore, it is difficult to determine what portion of rising electricity bills was caused by inflation, global conflict, broader supply-side pressure, or data center expansion. Still, several state-level reports have, at least in part, linked rising electricity costs to the growth of data centers. Some examples are summarized in Table 1.

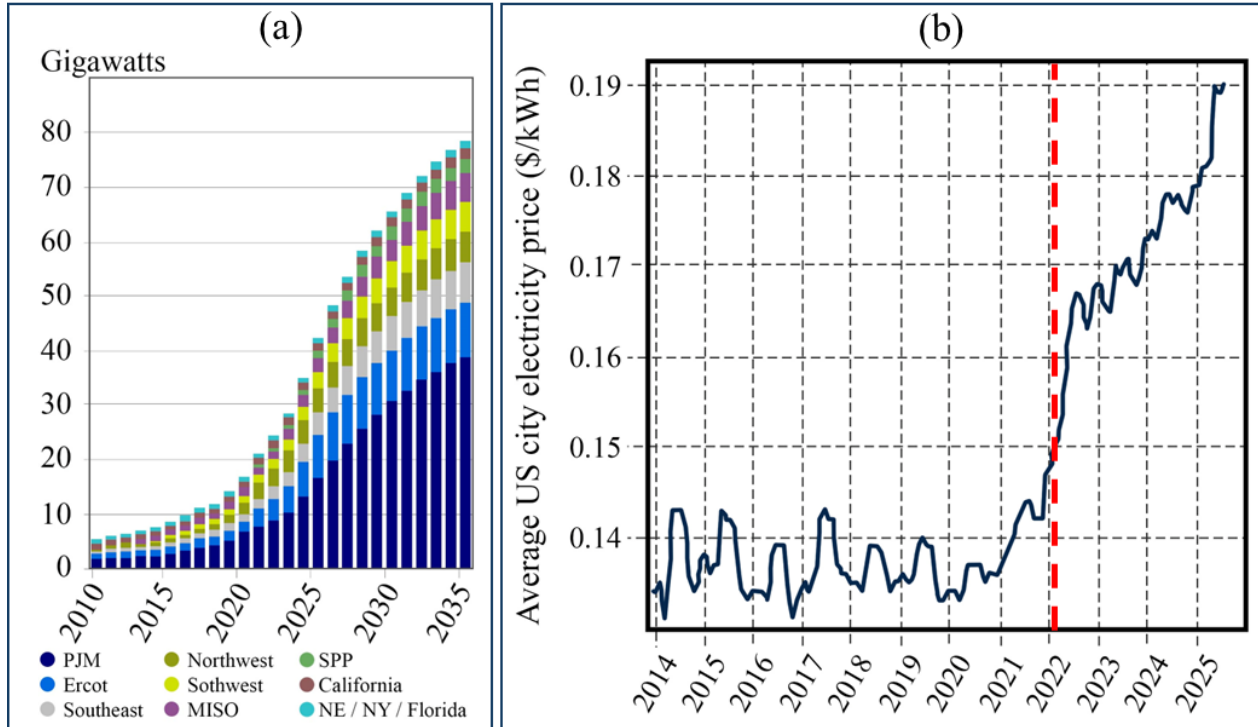


Fig. 6. (a) U.S. data centers power load; and (b) U.S. average electricity price (adapted from [10]).

Table 1. Examples of state-level reports linking rising electricity costs to data center expansion.

State	Document(s) relating electricity price increases to data centers
Arizona	Arizona Public Service (APS), the state's largest and longest-serving electric utility, has identified <i>data center growth as a key factor behind its increased electricity sales</i> , despite lower residential demand and energy-saving initiatives [10, 11].
Oregon	The Oregon Citizens' Utility Board (CUB) has stated that <i>the costs of serving data centers are being spread across all customers</i> , at a time when some electric bills in the state have risen by 50% over the past four years and utility disconnections have increased [10, 12].
Georgia	The Public Service Commission (PSC) and Georgia Power have pointed to rising global fuel costs, infrastructure upgrades, <i>and expanding demand (including electricity-hungry data centers) as key contributors to price increases</i> . Large data centers can require more than 100 Megawatts of electricity, enough to power roughly 80,000 U.S. homes [13].

Virginia	Data center industry is projected <i>to drive immense increase in energy demand. Building enough infrastructure to meet even half unconstrained data center demand remains challenging</i> , regardless of compliance with the Virginia Clean Economy Act (VCEA). Data centers have not yet increased residential electricity bills dramatically, largely because the state previously had unused energy capacity, but that reserve is now running out [14, 15].
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While on this topic, it is worth noting that several preliminary strategies could help reduce the energy impact of data centers. Three examples are briefly introduced here, with a more detailed discussion of potential solutions provided in Section 3.

- First, data center operations can become more flexible. Normally, only about 40% of the grid's total capacity is used on average because the system is designed to meet peak demands during the most extreme days of the year. If data centers reduce workloads during peak periods and invest in large-scale battery backup systems, it may be possible to add more capacity without requiring substantially new generation or transmission infrastructure [16].
- Second, co-location of data centers with new appropriately sized wind, solar, and energy storage projects can help reduce emissions, maintain reliability, and lower costs to some extent. However, this approach requires supportive state-level legislation and clean energy policies [17].
- Third, more transparent reporting of data center electricity demands can support long-term and early planning by utility companies for future load growth, clean energy needs, and cost allocations [17].

2.2. High Water Consumption

In addition to their high energy use, data centers also have high water consumption. Water is used for two main purposes in data centers. Firstly, it is used indirectly in the production of the energy required to operate data centers. Second, it is used directly for cooling the data centers facilities [2]. Since data centers operate under large and continuous computational workloads, they generate substantial heat and require heavy cooling. The most common data center cooling approach is evaporative cooling. Although alternative cooling methods (such as closed-loop cooling) also exist, they are still less common and associated with higher energy use [1].

A schematic representation of evaporative cooling in data centers is shown in Fig. 7. In this process, a cooling tower cools water through evaporation. The cooled water is then utilized in a heat exchanger to cool the warm air leaving the data center. Once the air is cooled, it is recirculated back into the data center to continue cooling the servers and other infrastructure. The cooling water is circulated through this process several times before being discharged and replaced with fresh water.

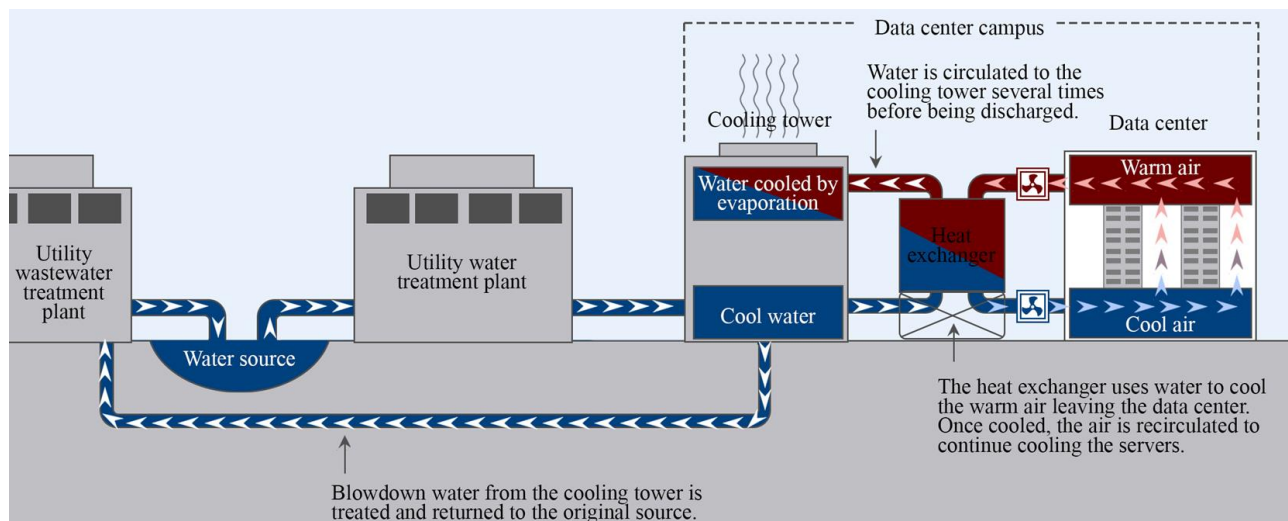


Fig. 7. Schematic illustration of evaporative cooling in data centers (adapted from [18]).

Facilities' water consumption is commonly calculated using the following relationship: $\text{water consumption} = \text{withdrawal} - \text{discharge}$. Based on this definition, residential buildings typically have a water consumption rate of only about 5% to 15%. This is relatively low since, for example, water used for showers is returned to the municipal wastewater system and may potentially be recycled. Notably, whether it is actually recycled depends on local infrastructure and planning, but it is considered non-consumed water in this type of calculation. In contrast, data centers have a much higher water consumption rate, typically around 50% to 80%, because a large portion of cooling water evaporates. (And the remainder is discharged to municipal wastewater facilities, which are already facing challenges in managing high volume of wastewater produced in cities.) Moreover, due to concerns related to corrosion and bacteria growth, approximately 80% to 90% of the water used for data center cooling is fresh drinking water [2]. These issues place data centers among the most water-intensive building types.

in 2023, U.S. data centers consumed 17 billion gallons of water directly for cooling, in addition to 211 billion gallons used indirectly through generating the electricity that powers them [2]. To put this in perspective, this is roughly equivalent to Chicago's annual water consumption. As shown in Fig. 8, this water consumption, driven largely by AI, is projected to increase to 2 to 4 times the 2023 level by 2028 [18].

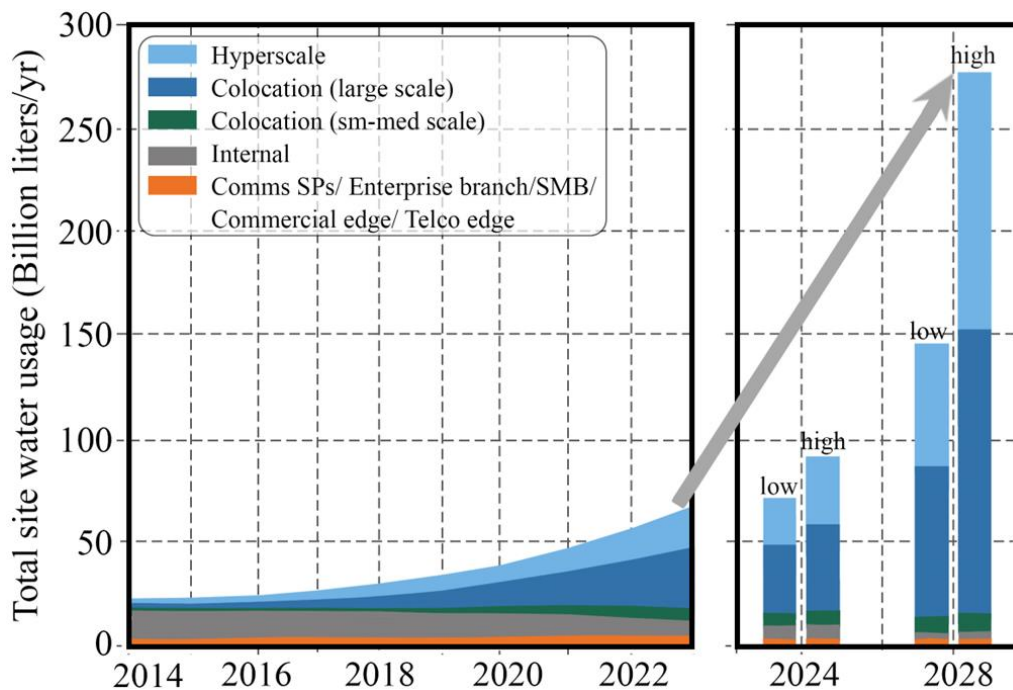


Fig. 8. Trend of U.S. data center water consumption (adapted from[18]).

More concerning, roughly 65% of new data centers built or under development since 2022 are located in areas facing high or extremely-high water stresses [19]. Fig. 9 shows the added data center capacities since 2022 in such areas. For example, a new data center in Newton County, Georgia consumes 500,000 gallons of water per day, which is equal to about 10% of the county's total water consumption. At the same time, additional data center permits are in progress, some of which would require up to 6 million gallons of water per day, more than doubling the county's current total water use [16].

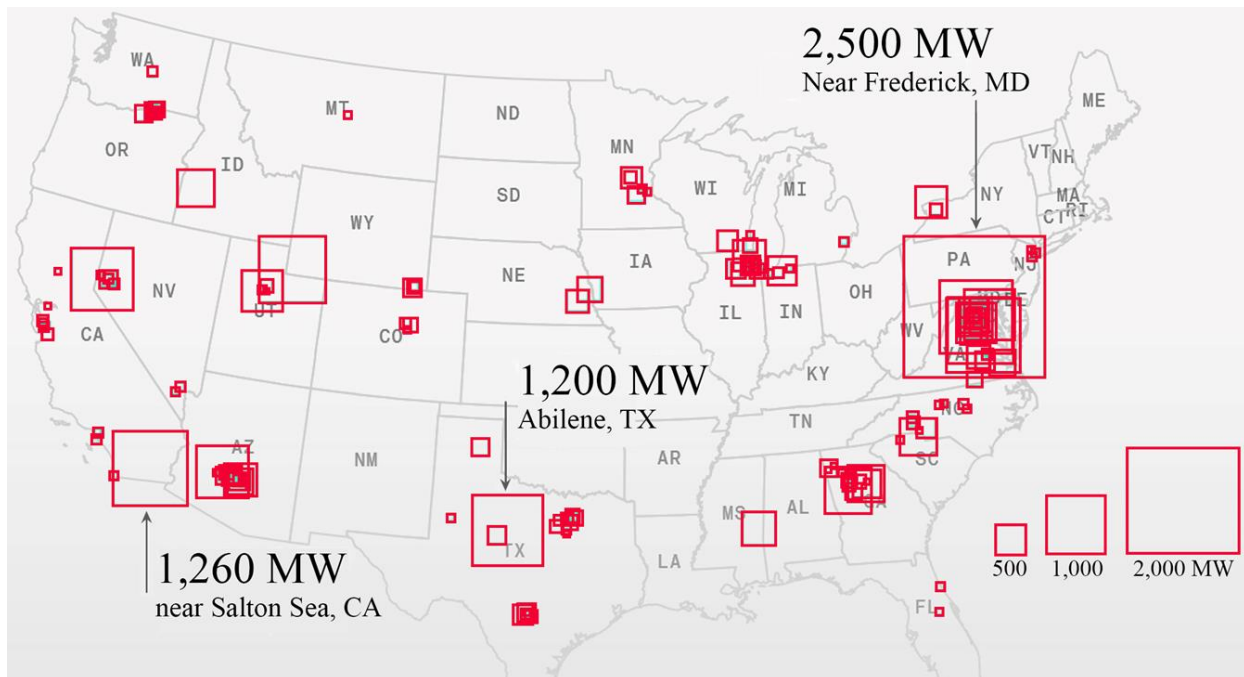


Fig. 9. Total new data center capacity in areas under high or extremely-high water stress (adapted from [19]).

2.3. Land Use and Environmental Concerns

The distribution of data centers across the U.S. is shown in Fig. 10. One noticeable trend in their location is that they are often built either constructed on what may be considered ‘best lands’ or in underrepresented communities, as explained below:

- a) **Best Lands:** Given the importance of uninterrupted uptime and high cost of AI infrastructure, minimizing natural disaster risk is a major consideration for data center providers. As a result, they are often drawn to locations that are more immune to natural disasters [5]. For instance, Northern Virginia, which hosts more data centers than any other location in the world, is strategically favorable in this regard because it faces near-zero earthquake risk, limited tornado risk, and lower hurricane risk than many surrounding states.
- b) **Underrepresented Communities:** Due to strong tax incentives, as well as lower costs for land, electricity, water, and construction labor, large clusters of data centers are increasingly being developed in areas with underrepresented communities [5], such as parts of Georgia and Alabama. For similar reasons, rural areas across the U.S. are also experiencing more land grabs for data center expansion.

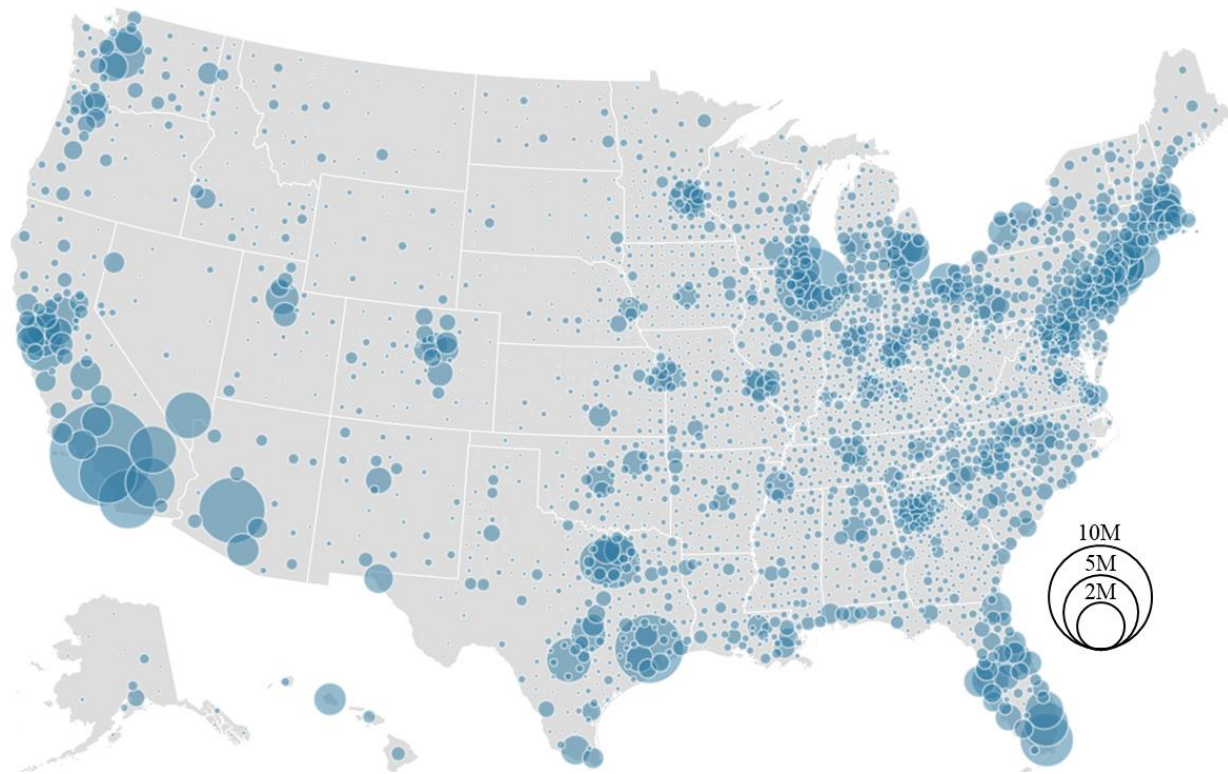


Fig. 10. Distribution of data centers across the U.S (adapted from [20]).

Although the construction of data centers can bring several benefits, if not properly regulated, it can also lead to significant environmental and community-related impacts, some of which are discussed below:

- a) **Large-Scale Environmental Damage:** For example, as shown in Fig. 11, a data center under development in Bessemer, Alabama is expected to clear 700 acres of woodland, despite strong public opposition. Public officials in Bessemer signed a non-disclosure agreement (NDA) related to this project, which limited the amount of information available to residents and contributed to substantial public concern. In addition, Alabama has no water quantity laws,

meaning that no state authority will evaluate whether the volume of water required by this project is reasonable or whether it could harm surrounding communities [21].

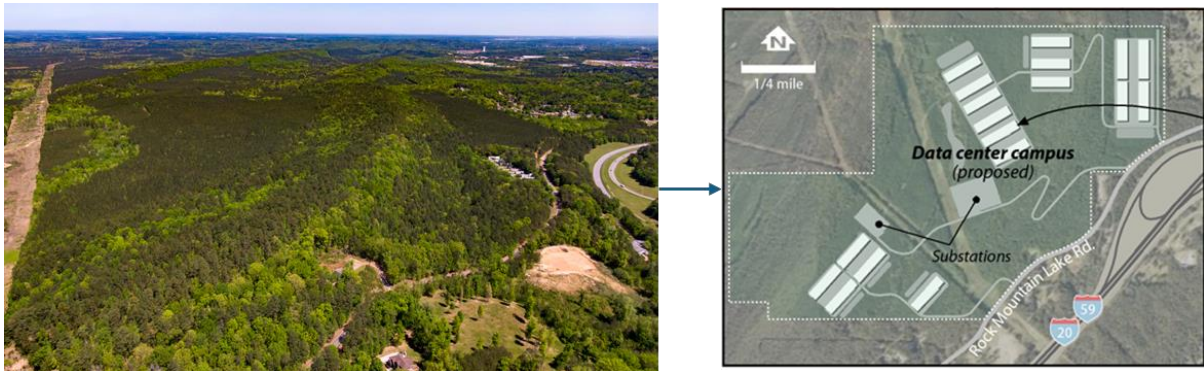


Fig. 11. A 4.5 million ft² data center project planned on 700 acres of woodland in Alabama (adapted from [21]).

- b) **Intolerable Human Disturbances:** In some cases, data centers are constructed only a short distance from residential neighborhoods. For example, several data centers in Loudoun County, Virginia (shown in Fig. 12) are located only a few meters away from residential areas, exposing residents to significant noise and pollution. The following quotes from long-term residents of the area illustrate the disruptive effects that unregulated data center development can have on nearby communities:

- “It was [...] beautiful, and now it has this very industrial feel across it. One can now drive for miles and just see data centers [22].”
- “I can no longer open the windows of my house on cool evenings because of the noise. Another neighbor put mattresses against their window to block the noise [22].”



Fig. 12. A data center located adjacent to residential homes in Virginia (adapted from [16]).

- c) **Farmland Takeover:** In some regions, farmland is being converted into data center sites, raising concerns about future food security and long-term local economy. The following quote is from a farmer in Las Vegas, Nevada, who sold a plot of farmland to a data center developer for more than one million dollars per acre, more than the land could have generated through agricultural use over a lifetime:

- “Piece by piece, [my family and friends] are selling their farmlands to a data-center developer. [I am] planning on moving out soon, the surroundings feel unfit for agriculture, or even for human life [21].”

Notably, these concerns are not limited to the United States. For instance, farmers in Singapore are also abandoning or selling their farmlands to major technology companies, raising similar concerns in other parts of the world [23].

2.4. Economic impacts

Data centers can bring several economic benefits to communities. These benefits may include temporary construction jobs, increased use of state renewable energy resources, and enhanced digital connectivity, which in turn can attract other businesses and support broader economic growth [24]. However, despite receiving substantial tax incentives and generating many temporary construction positions, data centers often create few permanent jobs and employ only 100 to 200 workers on a long-term basis [25]. A few examples are provided below:

- Since 2018, Washington State has spent approximately \$474 million in taxpayer funds to support data centers. An array of data center projects, with a total taxpayer cost of \$53.3 million, created only 260 jobs to meet the required threshold. This corresponds to an average public cost of about \$205,000 per job, or roughly three years of salary at the state’s median income [24].
- A 1.1 million ft² data center in Reno, Nevada is expected to create only 73 permanent jobs over the next decade [25].

These examples show that, although data centers can stimulate local economic activity, their long-term economic benefits may be more limited than often assumed. In many cases, substantial public incentives and tax breaks are provided, while the number of permanent jobs created remains relatively low. This raises concerns about whether the economic returns to local communities are proportionate to the public resources, land, and infrastructure committed to support data center development.

2.5. Health Impacts

Data centers contribute to a huge amount of air pollutants, such as CO₂ and NO₂. Typically, the pollution footprint of data centers is divided into three different scopes. *Scope 1* is direct CO₂ emission, which primarily comes from data center backup generators. This scope is normally low since, at least currently, backup generators of data centers are not turned on very often. *Scope 2* is CO₂ emission from the power grid that the data center is connected to. Normally lots of fossil fuels are burned in this scope to provide energy for data centers, contributing directly to multiple pollutions and harmful particles. This is the most substantial pollution source of data centers. *Scope 3* consists of remaining emission sources, such as the supply chain, construction, and transportation [18].

Although data center operators often collect environmental and health-related information for reporting to monitoring local authorities, they are generally not required to make this information publicly available [18]. As a result, the health impacts of data centers have not been studied or documented as extensively as other concerns, largely because of limited data availability. Nevertheless, some evidence indirectly links data centers to health-related consequences, with two examples described below.

- a) Fig. 13 shows part v of *Executive Order (EO) 14141: Advancing United States Leadership in Artificial Intelligence Infrastructure* [26]. This EO, signed by former President Biden,

prioritizes data center construction in areas where the cancer risk from air pollution is at or below the national average. Notably, this order was not terminated during the President Trump administration, suggesting that the health implications of data center expansion is among the issues that have received attention across different political administrations.

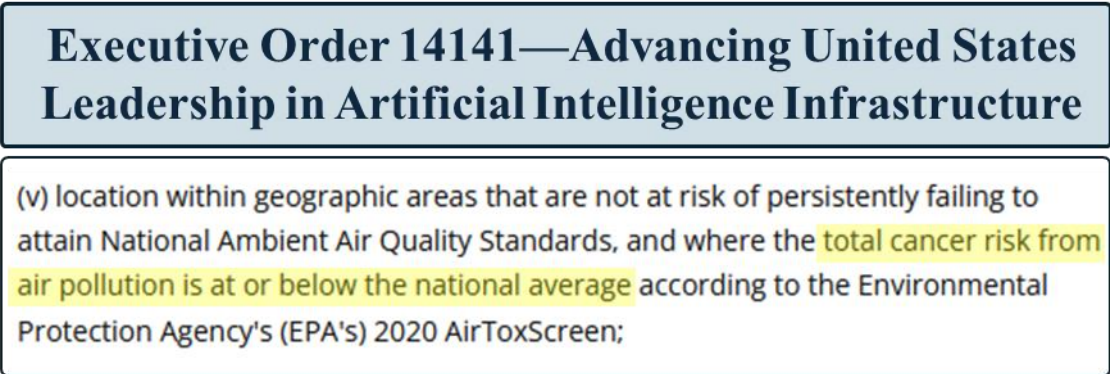


Fig. 13. Part v of Executive Order 14141, signed in January 2025 (adapted from [26]).

b) Fig. 14 (a) shows the distribution of health risk across several eastern states, where darker purple shades indicate higher health costs (in million dollars, normalized by population). The orange circles represent the locations of backup generators in Virginia, and the yellow circle marks the location of the generator cluster in data center alley (i.e., Loudoun County). Based on the underlying analysis associated with this figure, two key findings emerge. First, higher health risks are observed in areas near data center backup generators. Second, the most affected county is not Loudoun County itself, but Montgomery in Maryland, as also quantified in Fig. 14 (b). This is attributed to prevailing wind patterns that transport pollution from Loudoun to Montgomery County [18]. This raises an additional concern: the impacts of data centers may extend beyond the jurisdictions in which they are located, affecting populations in nearby areas, including those where local and state governments may be attempting to limit such impacts through stricter regulations.

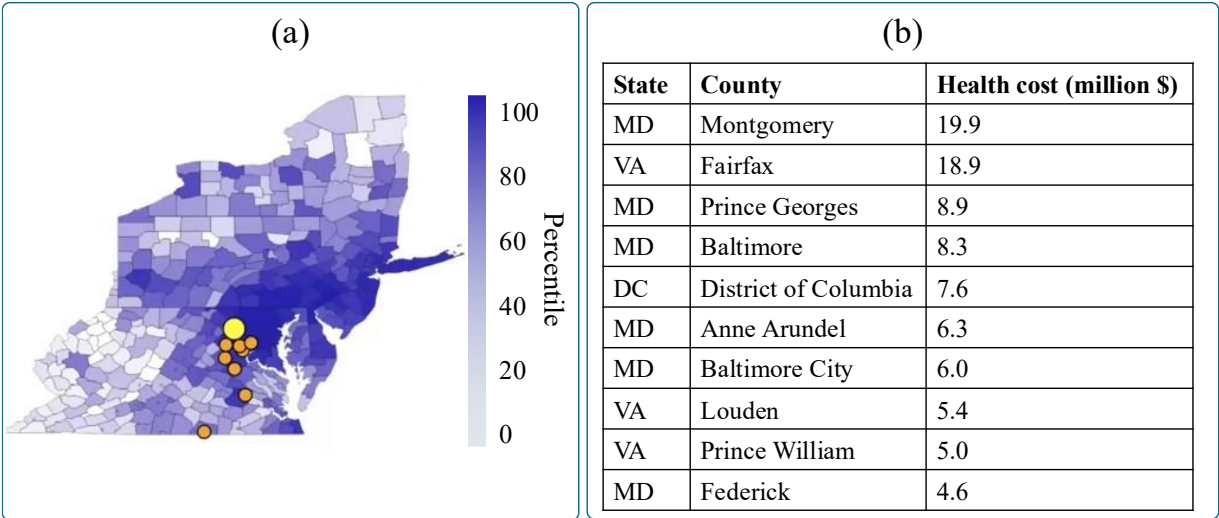


Fig. 14. a) Health cost map and the location of data center backup generators in Virginia; and b) top 10 eastern-states counties by health costs (adapted from [18]).

3. Regulations and Best Practices

Section 2 focused on identifying the potential negative consequences of data centers when their expansion is not adequately studied, planned, or regulated. To address these concerns, several solutions have been proposed. Some of the most impactful ones are outlined below.

3.1. Technical Innovation and Incentives

Similar to any rapidly growing technology, data centers require continued research, innovation, and investment to improve their energy efficiency and sustainability. Some of the most promising approaches currently being explored are summarized below.

- a) **Expanding Renewable Energy Integration:** This approach includes increasing investment in renewable energy, integrating data centers with renewable power sources, and making their operations more responsive to the availability and intensity of clean energy [1]. These methods can help reduce reliance on fossil-fuel-based electricity and lower the environmental impacts of data center operation.
- b) **Advancing Hardware and Algorithmic Efficiency:** This approach includes developing more efficient cooling technologies (e.g., advanced closed-loop water cooling systems with lower energy and water demands), as well as improving the efficiency of AI training and inference. Notably, AI models consume energy mainly in two stages [1]. The first is training, during which thousands of GPUs in a data center process large datasets to train AI model parameters. This stage can be extremely energy-intensive; for example, training a single AI model can emit as much carbon as five cars over their entire lifetimes [2]. The second stage is inference, in which trained AI models respond to user requests. Technological advancements in either training or inference can significantly reduce the energy demand of AI-driven data centers.
- c) **Enabling More Flexible Data Center Operation:** Examples include reducing power consumption during periods of peak residential demand, responding to real-time electricity prices, or co-locating with existing power plants to reduce the need for new infrastructure. Data center providers also have financial incentives to adopt such flexibility when market structures reward reduced or shifted electricity use. For example, Riot (a cryptocurrency mining company in Texas) used the flexibility of its operation in adapting to energy grid workload to reportedly save \$31.7 million. During some periods, the company earned more by providing flexibility to the grid than mining cryptocurrency itself [5]. Although cryptocurrency facilities differ from most data centers as their operations are more flexible and interruptible, this example still illustrates how operational flexibility can reduce grid stress and provide economic value.

3.2. Transparency

This section includes reducing the use of non-disclosure agreements (NDAs), informing the public in a timely manner about proposed data center developments, and publishing honest and accessible reports on energy efficiency and other potential impacts [5]. Improved transparency can reduce public fear, limit opposition caused by misunderstanding or lack of information, and help build trust between developers and communities. In addition, greater transparency with utility companies and other private entities can support earlier and more effective planning for

infrastructure upgrades and expansion, as well as more appropriate allocation of related costs, helping prevent those costs from being shifted to ordinary ratepayers [18].

3.3. Education and Awareness

Proper education of both the public and policymakers can support more responsible data center expansion. Public education can help people better understand both the benefits and potential impacts of data centers, as well as how to use AI tools more efficiently when possible. Educating politicians and decision-makers can also support better-informed policies, regulations, and planning decisions [1].

3.4. Regulatory Approaches

Regulation is one of the most critical requirements for responsible and controlled data center expansion. Clear and direct regulations are needed to ensure that utilities and data center providers pursue solutions that limit the need for unnecessary new infrastructure, avoid shifting costs disproportionately from data centers to ordinary rate payers, and support sustainable, energy-efficient, and responsible development. Because of the importance of this topic, this section presents examples of how different levels of government have begun developing rules and regulatory frameworks to protect their communities from the potential negative impacts of data centers.

3.4.1. Federal Level

In general, although the U.S. federal government has introduced several strong incentives, there still is no specific binding federal legislation designed exclusively for data centers. This largely reflects the country's federalist system, in which states hold significant regulatory authority, while federal mandates on how businesses operate often face political resistance. Two examples of these federal incentives are described below.

- a) U.S. Department of Energy (DOE) promotes data center energy efficiency through voluntary initiatives such as Data Center Energy Practitioner Program (DCEP), which provides training and recognition but does not establish binding targets. For example, the DCEP training program certifies energy practitioners who are qualified to evaluate energy performance and identify efficiency opportunities in data centers. Data centers can also receive voluntary certifications that benchmark their energy performance against industry peers using metrics such as PUE. However, these certifications do not carry legal or regulatory force [27].
- b) A more direct financial incentive is provided through the Inflation Reduction Act (IRA), which offer tax credits and tax reductions that lower the cost of greener technologies and practices. For example, operators may qualify for a 30% investment tax credits for on-site renewable generation, battery storage, or energy-efficient building systems. Since 2025, IRA incentives have shifted toward 'tech-neutral' credits, rewarding projects that achieve net-zero emissions regardless of the specific technology used [28].

3.4.2. State Level

Nearly all states have begun introducing regulations related to data center development and operations. Some of the most notable and relevant examples are included in this report. For clarity, these regulations are grouped into four main categories: energy, water, rates and accountability, and environment and health.

- a) Energy:

- California Building Energy Code, Title 24 (State: California)** [29]: This code establishes design requirements (e.g., hot aisle containment) intended to reduce data center cooling loads. New facilities that do not incorporate these efficiency measures at the design stage may not receive state approval. In addition, many data centers, especially colocation facilities built after 2016, have had to redesign or retrofit their layouts to comply with Title 24 compliance, reducing cooling loads by up to 20-30%. Notably, hot aisle containment is a common method for improving airflow and reducing energy consumption in computer rooms. As shown in Fig. 15, server racks are arranged so that their exhaust sides face each other, allowing hot air to be contained and removed without mixing with the cool supply air. This separation of hot and cold air improves cooling efficiency and supports reliable operation of critical infrastructure [30].

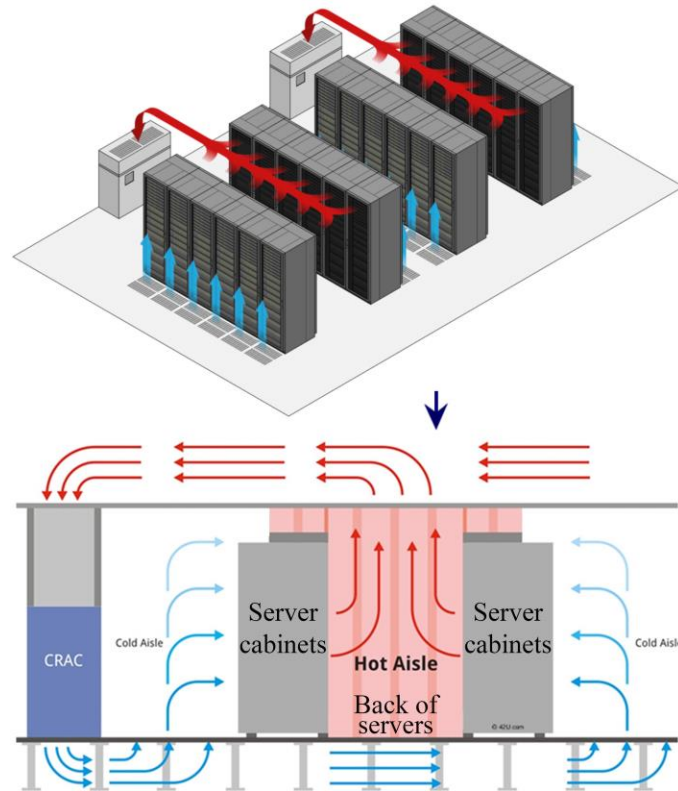


Fig. 15. Hot aisle containment design for more efficient cooling in data centers (adapted from [30, 31]).

Furthermore, all new data center developments since 2025 are required to use heat pumps for heating and cooling, as they are typically 3-4 times more efficient than traditional systems. These heat pumps must also meet specific efficiency thresholds, such as having the heating seasonal performance factor (HSPF) above 8.8.

- HF 16 (State: Minnesota)** [32]: This law requires data centers to obtain certification under a recognized green building standard, such as Leadership in Energy and Environmental Design (LEED), in order to qualify for sales tax exemptions on construction materials, electricity, and related equipment. Certification must be achieved within three years of the start of operations and must demonstrate compliance with benchmarks related to energy efficiency, sustainability, and operational performance.
- Senate Bill 6 (State: Texas)** [33]: Effective from December 2025, this law requires new large-load data centers to include remote disconnection capability, allowing the Electric

Reliability Council of Texas (ERCOT) to disconnect them during grid emergencies. This requirement reflects the growing pressure on the Texas grid, as the state has become one of the most attractive locations for data center development because of its abundant natural gas resources, strong wind and solar generation, available land, lower real-estate costs, and local tax incentives.

b) Water

- **Oregon Revised Statute (state: Oregon)** [34]: This Statute Limits water use for cooling systems, particularly in drought-prone regions of the state. Although it is not specific only to data centers, it applies to them as part of a broader group of large water users.
- **HF 16 (State: Minnesota)** [32]: In addition to its energy-related provisions, this law requires a separate and specific water-use permitting process for data centers, distinguishing them from other building types.

c) Rates and Accountability

- **Data Center Tariff DCT (State: Ohio)** [35]: This tariff requires data center customers to pay 85% of the electricity capacity they are subscribed to, regardless of whether they actually use that amount. The purpose is to prevent the cost of generation and transmission upgrades, which must often be made in advance to serve future data center loads, from being shifted to ordinary ratepayers if expected projects are delayed, downsized, or never fully materialize.
- **HB 3546 (State: Oregon)** [36]: This bill creates a separate category for data centers, cryptocurrency operations, and other large energy users, making them more directly accountable for the costs of their energy demand. Under this structure, these users must sign contracts for at least 10 years, pay for a minimum level of electricity use even if actual use is lower than expected, and pay additional charges if their demand exceeds agreed levels. This approach is intended to prevent excess costs from being passed on to households.
- **HB 2084 (State: Virginia)** [37]: This bill directs the State Corporation Commission (SCC) to review utility rates, tolls, charges, and customer classifications to determine whether data centers are categorized and charged fairly and reasonably, while maintaining equitable pricing structures.
- **Integrated Resource Plan (State: Georgia)** [38]: Approved in January 2025, this plan allows the state-regulated utility, Georgia Power, to charge data centers for electric service under terms intended to protect retail ratepayers from cost shifting. Specifically, new customers with more than 100 megawatts (MW) of demand should be billed under terms and conditions that differ from those applied to other customer classes, reflecting the risks associated with large electricity users. The plan also allows longer contract periods, increasing from 5 years to 15 years, as well as minimum billing requirements for high-load customers. These measures are intended to ensure that new large users do not shut down and leave the state before paying for the infrastructure built specifically to meet their demand.

d) Environment and Health

- **Clean Air Act Rule (State: Washington)** [39]: This rule applies to data centers that use emergency generators. It limits installation to no more than 21 emergency diesel engines, with each engine restricted to less than 4,423 brake horsepower (bhp).

- **Specific rule in Fairfax County, Virginia** [40]: Fairfax County requires equipment such as backup generators to be located more than 300 feet from residential areas because of concerns related to air pollution and noise. The county also requires mandatory noise studies to verify compliance with its Noise Ordinance.

4. Private Sector Commitments

In addition to federal initiatives and state regulations, the private sector (especially major technology companies driving much of recent data centers expansion) has also begun introducing its own commitments for more responsible data center development. One of the most detailed examples is *Microsoft's Community-First AI Infrastructure Plan*, launched in January 2026 [41]. As shown in Fig. 16, this initiative presents five main areas of commitment for building, owning, and operating data centers more responsibly within communities, as explain below.

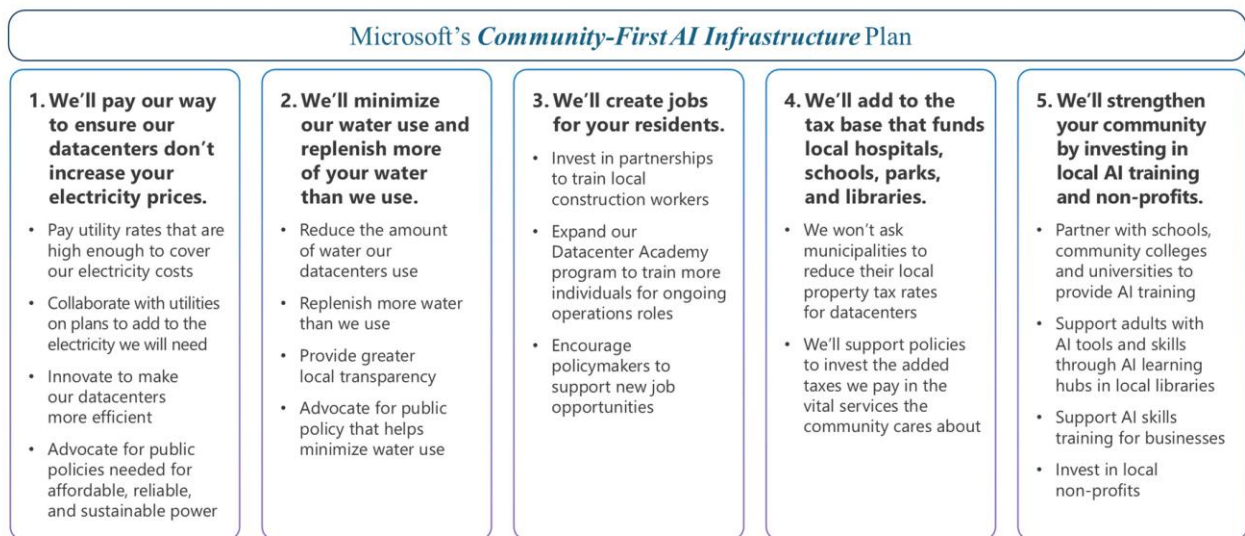


Fig. 16. Microsoft's Community-First AI Infrastructure Plan (adapted from [41]).

1) *Electricity*: Microsoft will ensure its data centers do not increase electricity prices for residential customers. Its approach includes paying the full cost of the electricity infrastructure needed to serve its facilities, coordinating early and transparently with utilities to support early grid expansion, improving the energy efficiency of its software and hardware, and advocating for policies that promote clean electricity generation, grid modernization, and community engagement. One example is Microsoft's contract to add 7.9 gigawatts (GW) of new electricity generation in MISO (a wholesale energy market that covers much of the Midwest), more than double its current consumption there.

2) *Water*: Microsoft commits to reducing its data center water-use intensity by 40% by 2030, replenishing more water than it withdraws, increasing transparency around water use, and supporting stronger public policy on water resilience. Its planned actions include expanding closed-loop cooling designs, reducing the use of potable water, partnering early with utilities to prepare water and wastewater infrastructure, and investing directly in community water systems. One example is its partnership in Quincy, Washington, where cooling water is treated and reused rather than relying on local groundwater. Another example is Microsoft's replenishment efforts such as leak-detection partnerships in the greater Phoenix area and wetland restoration projects in the Midwest.

3) *Jobs*: Microsoft also emphasizes local workforce development. The plan includes investment in trade training, apprenticeship programs, and expansion of the Datacenter Academy to prepare workers for construction, maintenance, and operations roles. For example, the Datacenter Academy partnership in Boydton, Virginia, has helped hundreds of students and adult learners earn industry-recognized certifications in information technology and critical facilities operations. Microsoft also considers future data center growth as a source of continued demand for skilled trades and related manufacturing jobs opportunities.

4) *Tax Benefits*: Microsoft will contribute to the local tax base rather than seeking reduced local property tax rates when establishing a data center presence. This is intended to help ensure that communities benefit financially through stronger support for services such as schools, healthcare, libraries, parks, and public safety.

5) *Community Investments*: This includes AI literacy and age-appropriate training programs for schools, colleges, libraries, and small businesses, as well as supporting local nonprofit organizations through employee volunteer and donation matching programs. One example is the AI bootcamp for students and faculty launched with Gateway Technical College in Mt. Pleasant, Wisconsin to cultivate a new generation of developers and creators of AI tools.

5. Summary and Future Roadmap

This report examined the potential consequences of adding data centers to communities without sufficient planning, investigation, and regulation. The main concerns were organized into five categories: energy, water, environmental, economic, and health impacts. Potential approaches for addressing these concerns, including transparency, education, technological advancement, and investment in cleaner energy, were then discussed. Next, examples of federal incentives and state-level regulations were reviewed to show how different jurisdictions are beginning to protect their communities while still allowing data center growth. Finally, selected examples of private-sector commitments were presented to illustrate how companies are also beginning to address community concerns.

Looking ahead, Pennsylvania is becoming an increasingly attractive location for large-scale data center development. Recent announcements include Amazon's plan to invest at least \$20 billion in AI infrastructure across the Commonwealth with first identified sites in Luzerne and Bucks Counties [42], a proposed \$6 billion CoreWeave data center project in Lancaster County [43], and a large multi-campus AI hub planned for Central Pennsylvania [44].

Although Pennsylvania lawmakers and state leaders have begun responding, the Commonwealth still does not have a fully established and comprehensive framework for protecting communities from the potential negative impacts of data center expansion. Several proposals are now in progress. For example, House Bill 1834 [45] (passed the Pennsylvania House in March 2026) would create a statewide regulatory framework involving energy, disclosure, backup generation, curtailment standards, and funding mechanisms. Additional bills and proposals have also emerged, including Senate Bill 724 [46], which would require data centers to pay for grid upgrades and enter community benefits agreements, and Governor Shapiro's proposed GRID standards [47], which call for developers to fund power, protect water, improve transparency, and hire local workers. However, these efforts are still developing, and they do not yet amount to a complete, settled statewide guardrail system.

For this reason, Pennsylvania should sooner move to define clear guardrails for responsible data center development before the pace of expansion outstrips the pace of policymaking. These

guardrails need to include protections against cost-shifting to residents, standards for energy sourcing and water use, noise requirements, transparency and community notification requirements, and mechanisms to ensure that local communities receive meaningful benefits from these projects. In this way, the Commonwealth can pursue AI and data center development in a way that is both innovative and responsible.

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